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GEOPHYSICS, VOL. XXIII, NO. 2 (APRIL, 1958), PP. 285–298, 9 FIGS.

SOME OBSERVATIONS ON ROCK MAGNETISM*

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ABSTRACT

It seems that in general the plane of maximum magnetic susceptibility lies in the bedding plane for sediments and in the plane of foliation for metamorphic rocks; there is, also, a tendency for the remanent vector to lie in the plane of foliation in the latter. In the case of chemical deposits, the question is raised as to whether the hematite crystal growth is controlled by the magnetic field. Since pure hematite crystals are paramagnetic along the ternary axis, the remanent vector lies in the basal plane perpendicular to this axis, which, being the plane of ferromagnetism, is also the plane of maximum susceptibility. We have investigated chemically deposited hematite in the Clinton iron ore of Silurian Age. Although the remanent vector lies close to the plane of maximum susceptibility, this plane, unfortunately, is also the bedding plane. Several other hematite-bearing formations show a direction of magnetization close to the bedding plane. Measurements of magnetization and susceptibility anisotropy of samples cooled below the transition temperature for hematite have been made with no conclusive results other than indications of the presence of hematite in some cases. Samples from the Hazel formation of pre-Cambrian Age have been investigated. The planes of maximum susceptibility for this slightly metamorphosed red bed dip at various angles, and thus a system of microfractures containing magnetic material is suggested as a possible explanation. Pole locations for the Clinton iron ore and the Hazel are presented.

INTRODUCTION

In recent years an increasing amount of work has been carried out on the problem of determining the direction of the earth's magnetic field at various times in the past by measuring the direction of remanent magnetism of rocks formed at these times. Quite a number of the measurements have been made with samples from red beds since hematite, which is thought to be the principal magnetic component of these beds and also to be responsible for their red color, has a high magnetic coercive force so that the rock tends to retain its early magnetization.

We (1956) have been interested in the magnetic properties of rocks containing hematite and especially in the mechanism by which hematite becomes magnetized in chemically deposited sediments; also Doell (1956) has posed the question of "crystallization magnetization." Furthermore, the rhombohedral hematite crystal has very interesting magnetic properties. Shull, Strauser, and Wollan (1951) have shown by neutron diffraction that the magnetic moments of the iron ions in a single crystal of hematite lie in parallel planes perpendicular to the ternary axis in the temperature range below the Curie point of 675°C. and above a transition temperature of about −20°C. Below the transition temperature, the moments turn into directions parallel to the ternary axis. In both states the ions are oppositely arrayed in an antiferromagnetic grouping. It has been found by Morin (1950) and Guillaud (1951), how-

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ever, that powdered hematite is weakly ferromagnetic in the first state (−20°C. to 675°C.) and paramagnetic in the second. In fact, Smith (1916) found in general that at ordinary temperatures a single crystal appears to be ferromagnetic in the plane perpendicular to the ternary axis but paramagnetic in a direction along the axis. On the other hand, Néel and Pauthenet (1952) and Haigh (1957) report that hematite possesses an isotropic component of ferromagnetism above and below the transition temperature in addition to the anisotropic component which disappears below the transition temperature. Chevallier (1951) demonstrated that as the crystal size of powdered hematite is reduced the particles lose their property of ferromagnetism and at a particle diameter of one micron, the powder is paramagnetic. At such a crystal size, the domain wall structure of the hematite should have disappeared. Recently, Li (1956) has suggested that the ferromagnetism of hematite arises in the domain wall structure.

With this background we may now raise several questions concerning the remanent magnetism of rocks bearing hematite. (1) Is the direction of the crystallographic axes of hematite controlled by the direction of the magnetic field when the mineral is deposited chemically? (2) If the orientation of the plane perpendicular to the ternary axis is oriented by the magnetic field or some other mechanism, is the remanent vector confined to this plane? (3) Is the plane or the line of maximum magnetic susceptibility of a red bed controlled by crystal structure of the hematite grains or by geometrical factors such as bedding laminations or orientation of elongate particles? (4) Can the presence of hematite be detected by measuring the magnetic properties of a rock at a temperature below the transition temperature? (5) Is the remanent magnetism of a red bed destroyed or disturbed when the outcrop is exposed to temperatures below −20°C.?

TECHNIQUE OF MEASUREMENTS

In measuring remanent magnetism of rocks we have used a spinner type magnetometer which is in a large part a duplication of the equipment described by Graham (1955) at the Department of Terrestrial Magnetism in Washington. The phase and magnitude of the signal induced in a pickup coil by the sample spinning at 270 RPS are measured, so that both direction and magnitude of magnetization can be deduced. The sample plugs are roughly one inch in diameter and 27/32 inch long.

The same equipment has also been used for making measurements of the anisotropy of magnetic susceptibility of rocks. The sample is now spun in a horizontal field at a half-speed of 135 RPS since the induced moment reverses in the sample once per revolution. Thus, a signal of 270 cps is generated, while the 135-cps signal from the remanent magnetism is removed by filtering. Therefore the change in susceptibility and direction of the maximum in the plane perpendicular to the axis can be measured. By spinning the sample about three mutually perpendicular axes, data are obtained for computing the directions of

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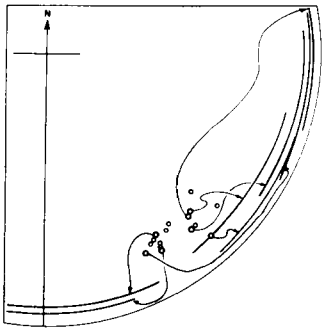


FIG. 1. Plot of magnetic vectors and traces of planes of maximum susceptibility for Clinton formation.

the principal axes of susceptibility and the differences in susceptibility between the directions of any two axes.

This method has some disadvantages over the coil and bridge type of measurement in that the susceptibility anisotropy effects cannot be separated from those due to electrical conductivity anisotropy. However, in most rock samples the conductivity is fairly low. A more disturbing disadvantage occurs when the remanence of the sample is sufficiently large that the second harmonic from its signal can interfere with the signal produced by the susceptibility anisotropy.

SILURIAN IN ALABAMA

Samples of hematite ore have been taken from the Clinton iron ore of Silurian Age in the Spaulding open pit mine of Republic Steel Company near Birmingham, Alabama. The formation, which dips at an angle of about 16° in a direction S 24° E., contains chemically deposited iron oxide and thus is of special interest. Both the remanent and the induced magnetism are high in these rocks. The remanent polarization amounts to roughly 10^{-4} cgs units per cc and the susceptibility to about 0.7×10^{-4} cgs units per cc.

Seven rock samples over a range of about 100 yards in a horizontal direction were taken, and Figure 1 shows a Schmidt-projection plot of the North-seeking vectors on the lower hemisphere, the downward direction being perpendicular to the bedding plane. Each circle represents the direction of the remanent magnetism for a plug, there being two plugs from each sample with one exception from which there were four plugs. Although our sampling does not cover as wide an area as is desirable to obtain a reliable pole location, it is interesting to compute the pole position, assuming the usual centered dipole field. It is also usually assumed that the axis of the dipole coincides on the average with the axis of spin of the earth at the time in question. The position of the earth's N magnetic pole is:

35° S. Lat. 42° W. Long.

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It is very interesting to compare this position with those we (1957) have found for other Paleozoic formations.

Carboniferous	(Barnett)	41° S. Lat.	52° W. Long.
	(Barnett)	42° S.	38° W. (South Pole)
Cambrian	(Sawatch)	49° S.	55° W.
	(Point Peak)	0°	22° W.

Hematite was present in thin sections obtained from both the Barnett and the Sawatch samples. A number of samples, obtained from calcareous concretions in the Barnett, were included in the measurements for the first mentioned pole position. The second position, which shows a reversed direction of magnetization, was obtained from limestone. The Sawatch samples consisted of sandy dolomite, while the Point Peak samples in general were siltstone. The data showed a large scatter in the case of the Point Peak. Thus, even though our data from the Sawatch and the Silurian are lacking in sampling areawise, the results, with the exception of the Point Peak, show a surprising clustering of the pole as found from samples of Paleozoic rocks of widely different age. We (1957) previously noted that the position of Carboniferous poles agree quite well with positions obtained by Belshé from Carboniferous samples in the British Isles. The positions of the Cambrian poles obtained from Point Peak samples also agree quite well with positions obtained by Creer from Cambrian samples in the British Isles. There is very good evidence, therefore, that there was a polar migration in Paleozoic times; such a migration has been pointed out by Runcorn (1955) and others.

We are faced here with the question as to whether the agreement in results we find from various Paleozoic formations happens by chance, is due to the magnetization of the Sawatch formation (Cambrian) and the Clinton iron ore (Silurian) taking place at times later than Silurian, or, perhaps, is due in some manner to the magnetic properties of hematite; finally, there is the possibility, for which there is little, if any, evidence from other investigators, that the poles did not move far in Paleozoic times. We (1957) have previously suggested the possibility of a late magnetization of the Sawatch, perhaps at a postdepositional time of dolomitization. In the case of the Clinton iron ore, the iron may have been oxidized or recrystallized to its present state long after deposition. This would not be incompatible with an original sedimentary origin. Since the position of the pole for this formation is intermediate between Cambrian (Point Peak) and Carboniferous (Barnett) positions, there is a possibility that the magnetization took place before Carboniferous time. Nevertheless, we should not overlook the possibility that this magnetization may represent the direction of the earth's field over a short segment of Silurian time, since the pole location is not too far from the expected position.

In Figure 1, we have also plotted segments of the traces of the planes of maximum susceptibility for plugs from each sample, the remanent vectors for these plugs being designated by heavy circles. It is readily apparent that the

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remanent vectors lie very close to the planes of maximum susceptibility and that these planes lie very close to the bedding plane of the formation.

To investigate the role of the hematite in the magnetization of these rocks, two samples were cooled in frozen CO_2 ($-78°$C.). Both of these plugs showed a drop in signal due to susceptibility anisotropy of about 20 percent. These experiments are fairly qualitative as the sample warms rapidly with spinning. One of these plugs, however, showed little if any change in remanent magnetism when cooled. On the other hand a natural crystal of hematite from Elba showed a decrease in remanence to about one-fourth its former value and also a change in direction when cooled. Upon warming of the crystal, the vector returned close to its former position but to about three-fourths its former intensity. Haigh (1957) has found that the near-saturation remanence for hematite powder decreases upon cooling below the transition temperature and increases upon warming above but not to its initial value. The high remanent magnetism of our crystal tends to interfere with the susceptibility measurements. Nevertheless, the anisotropy also seemed to decrease in amount and to change in direction. As would be expected for a single crystal of hematite, the remanent vector tends to remain in or near the plane of maximum susceptibility or of the plane perpendicular to the ternary axis. It should be mentioned that Smith found that crystals from Elba were not uniformly paramagnetic along the ternary axis. From the cooling experiments, in one case the decrease in anisotropy certainly indicated the presence of hematite in the Silurian sample while in the other, little if any indication of the presence of hematite is indicated by the lack of change of remanence. Since the plane of maximum susceptibility often coincides with the bedding plane in sediments, we can not decide whether the plane of maximum susceptibility is related to the magnetic field of the past and the remanent magnetization established by this field or to the bedding plane. The anisotropy of susceptibility could be due in large part to a layered structure of the hematite or other magnetic minerals instead of to a crystal alignment of hematite. In the case of detrital sediments, crystal alignment could be produced if the hematite particles in the settling process come to rest with a cleavage plane against the bottom surface.

TRIASSIC IN COLORADO

Figure 2 shows a plot of remanent vectors and traces of planes of maximum susceptibility from four plugs from three samples taken from red beds of the Lykins or the Gem formation, Triassic in Age. The formation dip is about 70° in a direction of S 60°W. The coordinates are with respect to the bed rotated to a horizontal position. The segments of the traces are plotted in the lower hemisphere and the vectors on the upper hemisphere.

Again, the remanent magnetism vectors lie fairly close to the planes of maximum susceptibility, although not so strikingly as in the previous example. The planes of maximum susceptibility are again very close to the bedding plane.

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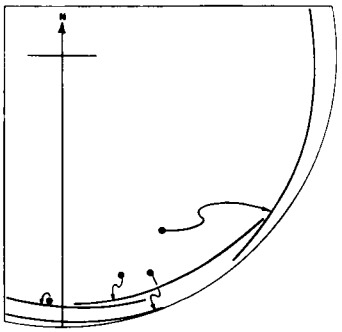


FIG. 2. Plot of magnetic vectors and traces of planes of maximum susceptibility for Lykins or Gem formation.

Upon cooling one of the plugs in frozen CO_2, the remanent magnetism decreased slightly in intensity. Thus again, the presence of hematite may be indicated, although the test is not definite since the cooling took place in the earth's magnetic field. We shall discuss this point later.

PRE-CAMBRIAN IN TEXAS

The Hazel formation composed of red beds of pre-Cambrian Age was sampled in Culbertson and Hudspeth Counties of West Texas. Figure 3 shows the Schmidt plot (lower hemisphere) of the remanent vectors for flat lying beds, each point representing an average of measurements from four plugs from a given sample of rock. Thus from five locations, scattered over about two miles, 15 samples were obtained.

Figure 4 shows plots for dipping beds, which have been rotated to a horizontal position in the diagram. The open circles are plots of the North-seeking vector on the lower hemisphere, and the solid circles are plots on the upper hemisphere. From nine locations, 39 samples have been obtained, the vectors being again averages obtained from four plugs from one rock sample. The formation dips

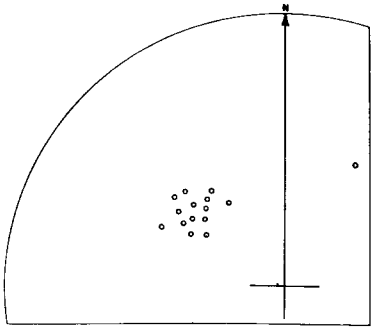


FIG. 3. Plot of magnetic vectors for flat lying Hazel beds.

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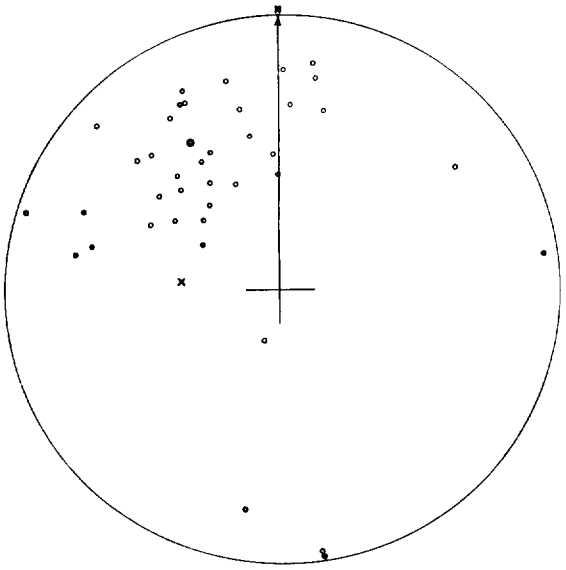


FIG. 4. Plot of magnetic vectors for dipping Hazel beds.

range as high as 85° and the samples are scattered over a distance of roughly 20 miles. It will be noticed that the average vector for the dipping beds designated by a heavy circle is less steep and somewhat closer to north but the scatter of points is very much greater. Both sets of measurements showed a slight scatter in the four vectors obtained from each sample. Also, after storage for a period of several months, some of the plugs showed a movement of vectors over a range of angles of about 2° to 16°, the plugs being stored with their vertical axes pointing both up and down. These variations are not severe.

The pole positions calculated from the two sets of data follow.

Flat Lying Beds

North pole 49° N. Lat. 175° W. Long. South pole 49° S. Lat. 5° E. Long.

Dipping Beds

North pole 60° N. Lat. 151° E. Long. South pole 60° S. Lat. 29° W. Long.

In Figure 4, the cross represents the average direction of remanence obtained by plotting the vectors in space with no account taken of dip. It is readily seen that if the magnetizations are moving more rapidly towards the direction of the present field in the dipping beds, the average vector would appear to be north of the average for the flat-lying beds. However, this picture is not clear-cut.

From measurements of Keweenawan rocks (pre-Cambrian) from the Lake Superior region, du Bois (1957) has located a line of polar migration for pre-Cambrian time which passes somewhat to the south of our North pole location obtained from the flat-lying Hazel samples. Since we do not know the age of the

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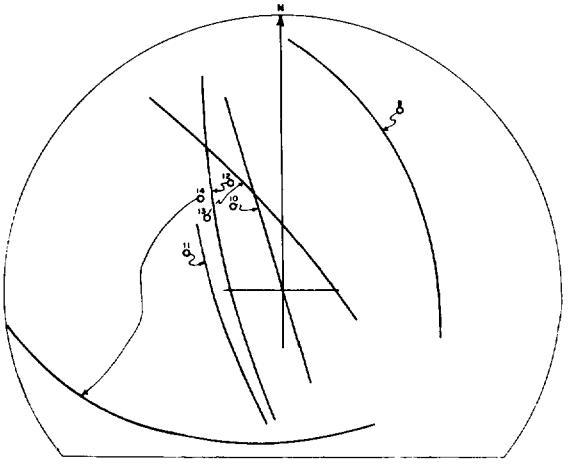


FIG. 5. Plot on lower hemisphere of magnetic vectors and traces of planes of maximum susceptibility for Hazel samples showing a strong linear direction of maximum susceptibility.

Hazel relative to the formations sampled by du Bois, it is difficult to make a comparison.

We have measured the anisotropy of susceptibility of a number of the samples, and in Figures 5 and 6 are shown the traces of the plane of maximum susceptibility along with the remanence vectors for the same plugs. Plots are made only for those plugs which show a maximum difference of susceptibility of over 3×10^{-6}. Furthermore, the difference in susceptibility between the maximum and minimum principal axes is more than twice the difference between

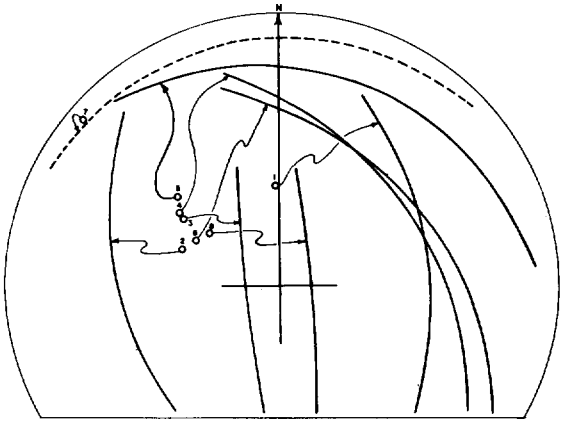


FIG. 6. Plot of magnetic vectors and traces of planes of maximum susceptibility for Hazel samples showing a strong planar direction of maximum susceptibility. All data are on lower hemisphere except dashed trace.

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the intermediate and minimum principal axes for the samples in Figure 5. The anisotropy in this case is fairly directional along the maximum principal axis. In Figure 6, the two differences are more nearly equal and thus the direction of maximum susceptibility tends to broaden out further into a plane. The total susceptibility of these samples is of the order 2 to 5×10^{-5}.

All of the plugs come from separate samples except 1 and 9, which are both very close to the dividing line of the two categories in the figures. Only one plug,

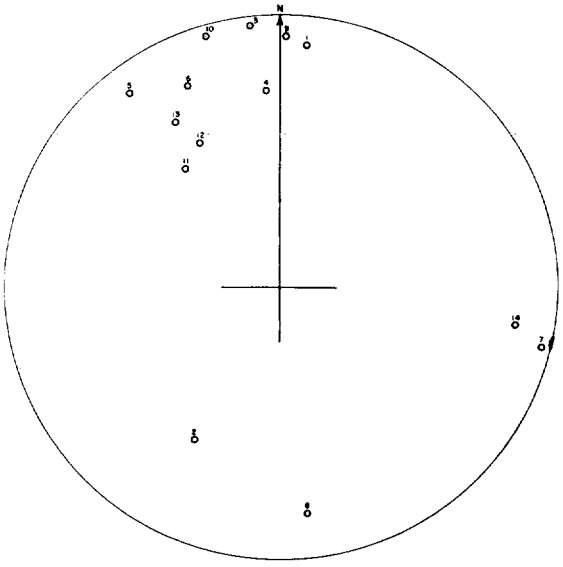


FIG. 7. Plot of directions of maximum susceptibility for Hazel samples.

7, comes from a dipping bed, the dip being 15°. The samples from the dipping beds seem to be lower in susceptibility than the samples selected. This result may be of no real significance.

The first striking observation is that the planes of maximum susceptibility no longer lie in the bedding planes. Curiously enough the remanent vectors seem to lie closer to the planes of maximum susceptibility in the plots of Figure 5 than in Figure 6 although the planes are more clearly defined in the latter.

Figure 7 shows a plot on the lower hemisphere of the directions of the maximum principal axis of these same plugs. Each of these points corresponds to an opposite point on the upper hemisphere since the principal axes have direction but not sense. The points cluster in the general neighborhood of the remanent vectors of the dipping beds but not of the flat-lying beds from which they are derived. Bhattacharya (1950) found a general correspondence between the remanent vector directions and the directions of maximum susceptibility of some samples of sedimentary rock. He concluded that elongate detrital magnetic

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particles might be aligned in the earth's field in settling and that the resulting direction of maximum susceptibility of the rock might be a more stable indicator of the direction of the earth's field than the remanent magnetism vector.

Such a picture is possible in our case. There is also another possible explanation, which may be more applicable. We might assume that the plane of maximum susceptibility corresponds to an average plane of microfractures in the rocks, in which fractures magnetic minerals are deposited. It will be noticed in Figures 5 and 6 that the planes dip at various angles to the horizon but in general intersect the horizon in a line which runs from slightly East of South to slightly West of North. Thus, if in the same sample, fractures filled with magnetic mineral dip at various angles but have a common line of intersection, this line would be expected to be a direction of maximum susceptibility. The Hazel has been described by King and Flawn (1953) as being partly metamorphosed; however, it is not foliated. Even the bedding planes are detected with difficulty. We shall discuss metamorphic rocks in a following section.

Along these lines Graham (1954) has reported finding a maximum susceptibility in the bedding planes of sediments with the exception of a folded formation in which the maximum direction was perpendicular to the bedding plane. He also found the maximum susceptibility to be parallel to the lineation in the case of two schists.

Balsley and Buddington (1957) have found the direction of remanence to be at right angles to the lineation in metamorphic samples containing magnetite. As for metamorphic rocks containing hematite, they have suggested the possibility of the anisotropy being a controlling factor in magnetization in the case of oriented crystal grains.

When plugs 1 through 14, as designated in Figures 5, 6 and 7, were cooled in carbon dioxide snow, the remanent magnetizations were changed to new directions in a rather strange pattern; some increased in intensity and some decreased. Upon warming of the plugs, tests indicated that the magnetizations returned close to their former directions and intensities. After further experiments with plugs 2 and 3, it was found that the changes depended on orientation of the sample in the earth's field during the cooling. In the case of both the single crystal from Elba and a sample of hematitic ore from Alabama obtained from Ward's Natural Science Establishment, the magnetization decreased after cooling, even though the earth's field aided in direction during the cooling. There seems, therefore, to be a different effect present in the case of the Hazel samples. The most likely explanation is that the magnetic relaxation time predicted by Néel's (1955) theory becomes sufficiently long for some of the fine magnetic particles at the low temperature, that these particles retain for a time the magnetization produced by the earth's field. The tests should thus be made in a field-free space.

The susceptibility anisotropy dropped in magnitude when plug 3 was cooled, and thus the presence of hematite was indicated.

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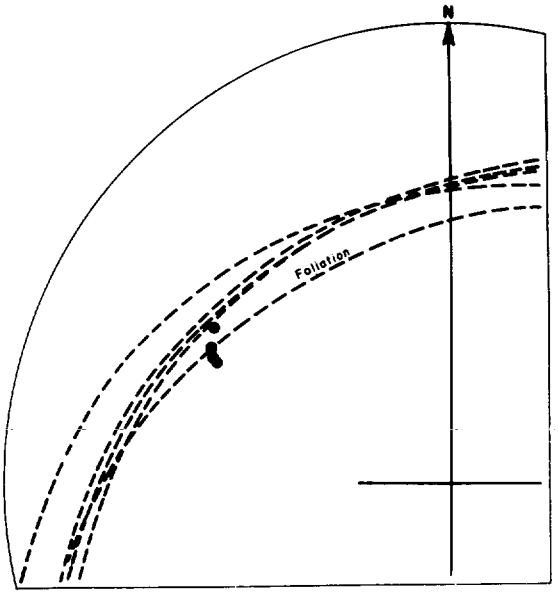


FIG. 8. Plot of magnetic vectors and traces of planes of maximum susceptibility and plane of foliation for Valley Springs Gneiss.

METAMORPHIC ROCKS

Several years ago, Mr. P. A. A. H. Masson pointed out that the remanent magnetism vector seemed to lie in the plane of foliation in the case of several samples of metamorphic rock obtained from a well core. We have more recently made measurements of both remanence and susceptibility anisotropy in metamorphic rocks. Figure 8 shows a plot on the upper hemisphere of results obtained from four plugs from the same rock sample of Valley Springs Gneiss outcropping near Burnet, Texas. Here, the remanent vectors are very close in position to the trace of the plane of foliation and to the planes of maximum susceptibility. Figure 9 shows a plot on the upper hemisphere of results obtained from two plugs, each cut from a separate rock sample of Packsaddle Schist outcropping in the Llano uplift area of Texas. Again the vectors lie quite close to the planes of maximum susceptibility and to the plane of foliation. Both of these formations showed strong anisotropies, the maximum difference in susceptibility amounting to roughly one-third to one-half of the total susceptibility.

Kalashnikov, Petrova, and Grabovsky (1957) recently have mentioned finding magnetic permeabilities in metamorphic rocks an order or two greater along "the layers" than across. The remanence behaves in a similar manner. They also report that the remanence seems more stable across "the layers" than along. This type of stability and the magnitude of the permeability anisotropy are surprising.

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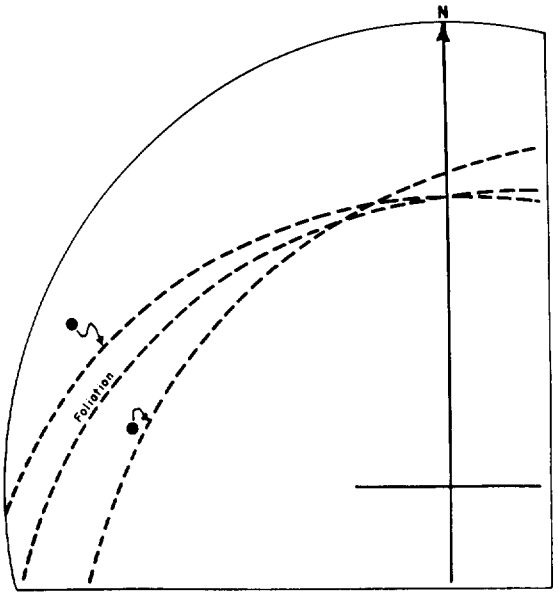


FIG. 9. Plot of magnetic vectors and traces of planes of maximum susceptibility and plane of foliation for Packsaddle Schist.

When a plug of the Valley Springs Gneiss was cooled in carbon dioxide snow, the anisotropy signal increased when the sample was spun about two axes but decreased about the third. There may thus be hematite present. A sample of the Packsaddle Schist showed no change in anisotropy after cooling. The anisotropy in this case is probably due in the main to the geometry of the planes of foliation.

SIDERITE

In connection with making measurements on rock magnetism, it might be mentioned that although siderite (Fe_2CO_3) is paramagnetic, its susceptibility is fairly high, the values of samples measured by us ranging from roughly 2 to 4×10^{-4}. The maximum difference of susceptibility due to anisotropy ranges from roughly 1/20 to $\frac{1}{2}$ of the total. In the case of a siderite from Hill County, Texas, the plane of maximum susceptibility seems to coincide with the bedding plane. In the case of a nearby siderite concretion, the direction of maximum susceptibility seems to be perpendicular to the bedding plane. A sample from a calcareous silty shale concretion in the Pierre shale in Colorado also showed a maximum susceptibility in the vertical direction; the bed dips about 26°. Again, the fracture system may play an important role in the anisotropy. It should be mentioned that in making spinning measurements of remanence with samples of high anisotropy, the vertical component of the earth's field should

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be cancelled. Since the vector induced by the vertical component of the earth's field is not vertical due to the anisotropy, there is a horizontal component which does not reverse in the sample as it spins. An unwanted signal is thus set up.

CONCLUSIONS

It appears that, in general, undisturbed sediments show a maximum susceptibility in the bedding plane. Again, metamorphic rocks show a maximum in the plane of foliation and undoubtedly in many cases are magnetized in a direction lying in this plane.

In the case of the Hazel formation which is slightly metamorphosed with no foliation, the direction of the plane of maximum susceptibility varies. It is suggested that microfractures filled with magnetic minerals may account for this pattern of anisotropy.

Regarding the question of hematite crystals forming with a relationship between the magnetic field and the basal plane perpendicular to the ternary axis, we have no conclusive evidence. In the Clinton iron ore formation containing chemically deposited iron oxide, although the remanent vector lies close to the plane of maximum susceptibility, unfortunately this plane also lies close to the bedding plane. Perhaps x-ray analysis might throw light on this problem.

As to detecting hematite by measuring the change in remanent magnetism and susceptibility anisotropy of a sample cooled below the transition temperature, the method has the advantage, over heating above the Curie point, that it does not produce the marked chemical and crystallographic changes caused by heating. Other magnetic minerals, furthermore, do not possess corresponding transition temperatures so far as is known; on the other hand, both hematite and magnetite have Curie points separated by only about 100°C.

As to the problem of outcrops subject to low temperatures in frigid climates, we find little if any evidence of serious change in remanent magnetism for hematite-bearing rocks exposed to such temperatures.

In conclusion, we feel that studies of rock magnetism will be of great value not only in the broad problems of polar wandering and continental drift but also in specific problems such as dating, postdepositional changes in sediments and perhaps in some cases, investigations of regional stress patterns.

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